Yongchun Li - Week 6 Report (*1378156*)

1. Tasks Accomplished This Week

Meetings & Collaboration:

1. Attended daily stand-up meetings (1 h).
2. Participated in the weekly team retro & next week planning(2 h).

Development Work: see commit link

3. Implemented comprehensive backend utility classes from scratch (~2 h):
   - ✓ Built StringUtils, Convert, ServletUtils, IpUtils, and CharsetKit classes
   - ✓ Added detailed English documentation for all utilities
   - ✓ Standardized error handling and fallback mechanisms across most utilities
4. Developed full-featured logging system with visualization improvements (~1 h):
   - ✓ Created RequestLoggingFilter for HTTP request/response tracking
   - ✓ Implemented LogAspect with AOP for method-level logging
   - ✓ Added colorized console output with ANSI formatting for improved readability
   - ✓ Integrated visual indicators for performance metrics

2. Key Issues Encountered

Initial implementation had some redundancy between filter-based and aspect-based logging, which was resolved by establishing a shared visual style and formatting conventions.

ANSI color code compatibility required testing across different terminal environments. Some initial implementations caused rendering issues in certain IDEs, which required adjustments to ensure wide compatibility while maintaining visual enhancements.

The stringutils class has inheritance differences with Yun Chen's implementation, and may need further merging

3. Plan for Week 5

Sprint 2 – Remaining Responsibilities:
Begin implementation of authentication and authorization system
Write unit tests for utility classes and logging components
Continue refining system architecture Diagram and db design.
Continue to add needed functions to utils.